9 What is the gravitational potential at O and its direction?

	potential	direction
A	0	no direction
B	$-\frac{GM}{r}$	towards Z
C	$\frac{GM}{r}$	towards Z
D	$-\frac{4GM}{r}$	no direction

Q9. Ans: D

Gravitational potential is a **scalar** quantity, so we simply add the contributions from all three planets without worrying about directions.

$$\phi = -\frac{GM}{r} - \frac{GM}{r} - \frac{G(2M)}{r}$$

$$\phi = -\frac{4GM}{r}$$

Gravitational potential is always **negative** and is a scalar — it has **no direction**. Looking at the options, option D gives magnitude $\frac{4GM}{r}$ with no direction, which matches (the negative sign is implied by the nature of gravitational potential).